



kingdom of Saudi Arabia
Ministry of Education
Hafr Al Baten Directorate for boys
Al –Ajial International Schools



Student's Name		Grade	9
Subject	Computer	worksheet	

Q1: choose the correct answer:

1- Which algorithm is used for clustering?

Linear Regression	K-Means	Decision Tree	Data
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2- Which type of learning uses labeled data to train models?

Unsupervised	Reinforcement	Semi-supervised	Supervised
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3- Which technology enables AR devices to map environments in real-time?

SLAM	GPS	RFID	NFC
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4- Which tool is commonly used for developing AR applications on iOS?

ARCore	Unity	ARKit	Unreal Engine
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Q2: Read the following and put (✓) for the correct answer and (x) for the wrong answer

1) Unsupervised learning requires labeled data.	(x)
2) Feature engineering involves selecting relevant input variables.	(✓)
3) A confusion matrix is used only in regression tasks.	(x)
4) Sentiment analysis classifies text emotion.	(✓)



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Q1: choose the correct answer:

1. What is the primary purpose of animating game objects?

- A) To increase the game's file size
- B) To make objects static and unchanging
- C) To create movement and bring objects to life
- D) To remove interactivity from the game

Answer: C) To create movement and bring objects to life

2. Which of the following is a common technique for 2D animation in games?

- A) Rigidbody physics
- B) Sprite sheets
- C) Mesh deformation
- D) Raycasting

Answer: B) Sprite sheets

3. In Unity, what component is used to control animations for a game object?

- A) Collider
- B) Animator
- C) Rigidbody
- D) Renderer

Answer: B) Animator



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Q1(A) Match the word in column (A) with the correct option
in column (B):

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Attribute	#	value
1- NER	2	Represents words as vectors
2- word2vec	3	Evaluates machine translation
3- BLEU Score	4	Identifies person/organization names
4- Syntax Tree	1	Parsing sentence structure

(B) Re-arrange the next stages to the ML pipeline:

- (3) Train model
- (2) Preprocess data
- (1) Collect data
- (4) Evaluate model



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Q1: complete the following from between brackets:

(**SMOTE** - **Overfitting** - **Reinforcement** - **Features**)

1- The input variables in a dataset are called **Features**

Reinforcement

2- _____ learning uses rewards to train models.

3- A model with high variance is prone to **Overfitting**

SMOTE

4- _____ is a technique to handle imbalanced datasets.



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Q1: choose the correct answer:

1- Which file format is commonly used for skeletal animations in 3D games?

- A) .PNG
- B) .FBX
- C) .MP3
- D) .TXT

Answer: B) .FBX

2- Which of the following is NOT a common animation blending technique?

- A) Linear blending
- B) Additive blending
- C) Multiplicative blending
- D) Overwrite blending

Answer: C) Multiplicative blending

Short Answer Questions:

1. What is the purpose of an Animator Controller in Unity?

Answer: The Animator Controller manages animation states and transitions using a state machine, controlling which animations play under specific conditions.

2. What is a keyframe in animation?

Answer: A keyframe defines the start or end point of a smooth transition, storing an object's properties (position, rotation, scale) at a specific time.



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Q1 : Read and put True(✓) or false (✗) :

1. Animation in games involves changing the position, rotation, or scale of objects over time.
○ Answer: True
2. Keyframes are used in animation to define the start and end points of an object's movement.
○ Answer: True
3. In Unity, the Animator component is not required to play animations on a GameObject.
○ Answer: False
4. Sprite sheets are used only for 3D model animations, not 2D animations.
○ Answer: False
5. Inverse Kinematics (IK) is a technique used to make character animations more realistic by adjusting limb positions based on the environment.
○ Answer: True